

MULTI-TASKING REMINDER GADGET With Electronic Calendar

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This multitasking gadget supports elderly patients with memory impairments, individuals with infectious diseases, and students who need help staying organised. Equipped with a real-time clock, it functions year-round as an electronic calendar, accurately displaying the day, date, time, and year.

It reinforces daily routines by issuing timely reminders for medication, study sessions, homework, playtime, and bedtime. If an alarm is ignored, a message is sent to a designated mobile phone, prompting necessary action.

Required components are listed in Bill of Materials table, while Fig. 1 presents the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of the multitasking reminder and electronic calendar. It is built around an Arduino Uno (MOD1), RTC3231 (MOD2), SIM800L GSM module (MOD3), a 16x2 LCD I2C module (MOD4), a buzzer, and a few other components.

The key components include:

- Arduino Uno: Used for programming
- LCD display: Shows reminder details
- GSM module: Sends and receives messages and calls

The gadget ensures that patients or elderly individuals take their medication on time by providing timely reminders. For example, if someone needs to take Crocin at 2:00 PM, the alarm will go off precisely

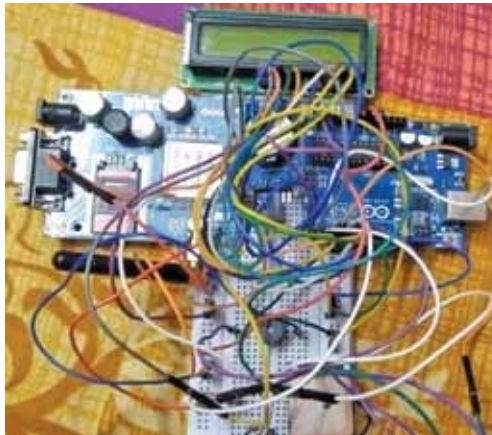


Fig. 1: Author's prototype

BILL OF MATERIALS

Name	Quantity
Arduino Uno (MOD1)	1
RTC3231 (MOD2)	1
SIM900A/SIM800I GSM module (MOD3)	1
16x2 LCD I2C module (MOD4)	1
Buzzer (B1)	1
LED (D1)	1
Push-to-on switch (S1)	1
330Ω resistor	1
12V, 2A adaptor	1

at 2:00 PM, displaying the medicine name and dosage (such as one tablet or a spoon of syrup) on the screen.

If the patient does not turn off the alarm by pressing the switch (S1), the alarm will restart after three seconds, and the medicine details will reappear on the display. If the patient still does not respond, an LED light on the gadget will turn on, and a message will be sent via the GSM module to the patient's guardians (such as parents or caregivers),

notifying them that the medicine has not been taken. This allows them to follow up and ensure the patient follows the prescribed schedule.

Software

The code is developed using the Arduino IDE. Install the required libraries via the Library Manager to set up the environment. Navigate to: Sketch→Include Library→Manage Libraries

Then, search for and install the following libraries:

- LiquidCrystal
- LiquidCrystal_I2C
- RTCLib

First, configure the type of LCD being used. If using an SPI LCD, comment out the I2C LCD line and uncomment the SPI LCD section. By default, the code is set for an I2C LCD. Next, define the other libraries and pin configurations in the code.

All necessary operations—such as checking button status, updating the date and time, and handling other device functions—are executed in the loop function. The sendSMS function allows users to set message content and recipient phone numbers in the code.

Now, select Arduino Uno as the board and choose the correct port number based on the laptop connection.

Finally, upload the code.

Construction and testing

After uploading the source code to the Arduino Uno, disconnect the USB